

Topología y direccionamiento

Equipo	Interfaz	IP	Rol
PC0	—	192.168.0.10/24 (GW 192.168.0.1)	interna
PC1	—	192.168.0.12/24 (GW 192.168.0.1)	interna
MiWeb	—	192.168.0.15/24 (GW 192.168.0.1)	servidor web interno
Router0 (borde)	Gi0/0	192.168.0.1/24	NAT inside
Router0 (borde)	Gi0/1	10.0.0.2/30	NAT outside
Router1 (ISP)	Gi0/1	10.0.0.1/30	enlace con Router0
Router1 (ISP)	Gi0/0	64.223.190.1/24	hacia Google
Google	—	64.223.190.94/24	servidor Internet

Bloque público asignado: 200.5.224.0/24 (solo usado como direcciones de traducción). MiWeb se publica en 200.5.224.50; las PCs usan un pool de 2 IPs (200.5.224.100–101) con overload.

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Technical Support: http://www.cisco.com/techsupport
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Image text-base: 0x2100F918, data-base: 0x24729040

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Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

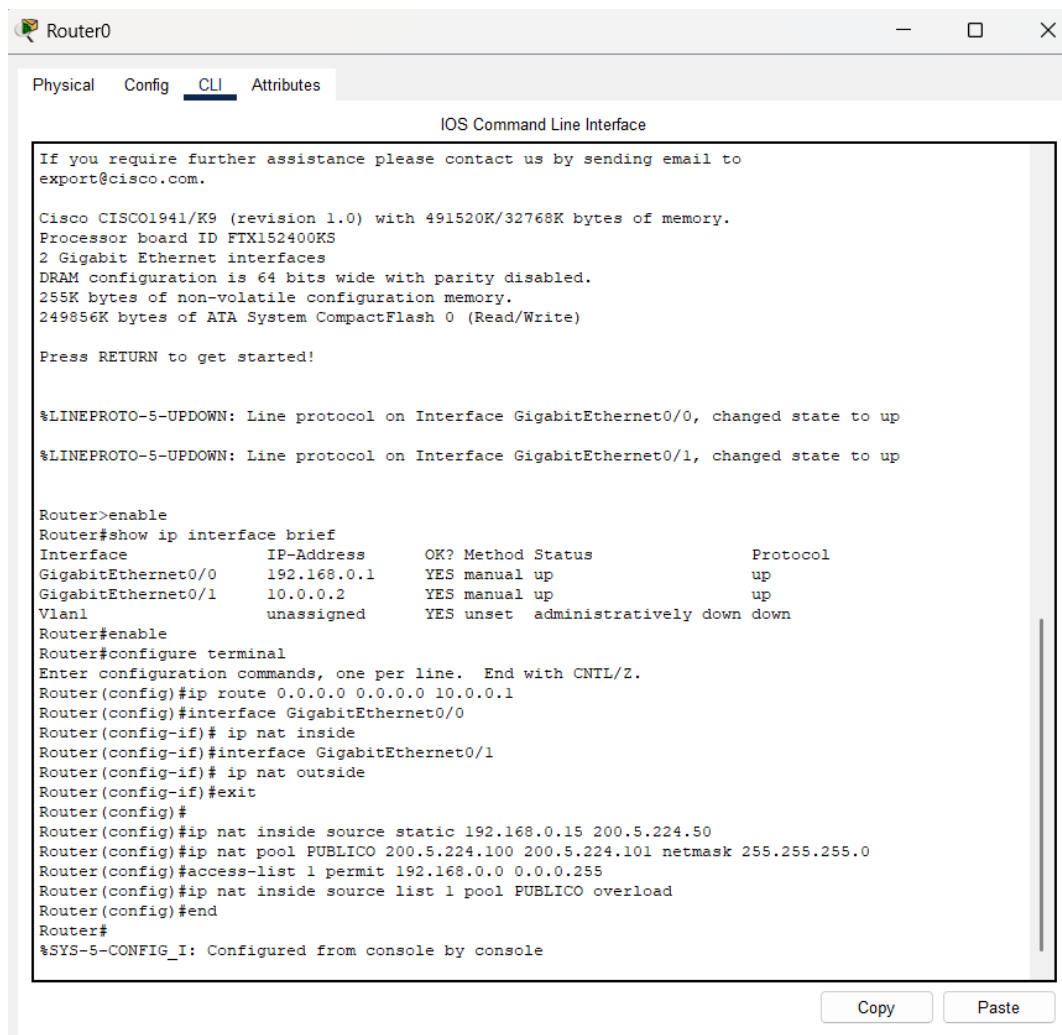
Router>enable
Router#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0  192.168.0.1     YES manual up              up
GigabitEthernet0/1  10.0.0.2        YES manual up              up
Vlan1              unassigned      YES unset  administratively down down
Router#
```

Copy Paste

Router0: Gi0/0 = 192.168.0.1 (inside), Gi0/1 = 10.0.0.2 (outside).

Configuración de NAT en Router0

```
ip route 0.0.0.0 0.0.0.0 10.0.0.1
interface GigabitEthernet0/0
 ip nat inside
interface GigabitEthernet0/1
 ip nat outside
! NAT estatico (MiWeb)
ip nat inside source static 192.168.0.15 200.5.224.50
! NAT dinamico con overload (pool de 2 IPs)
ip nat pool PUBLICO 200.5.224.100 200.5.224.101 netmask 255.255.255.0
access-list 1 permit 192.168.0.0 0.0.0.255
ip nat inside source list 1 pool PUBLICO overload
```



The screenshot shows the Router0 CLI interface with the following content:

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

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%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
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Router>enable
Router#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0  192.168.0.1     YES manual up              up
GigabitEthernet0/1  10.0.0.2        YES manual up              up
Vlan1              unassigned      YES unset  administratively down down

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.1
Router(config)#interface GigabitEthernet0/0
Router(config-if)# ip nat inside
Router(config-if)#interface GigabitEthernet0/1
Router(config-if)# ip nat outside
Router(config-if)#exit
Router(config)#
Router(config)#ip nat inside source static 192.168.0.15 200.5.224.50
Router(config)#ip nat pool PUBLICO 200.5.224.100 200.5.224.101 netmask 255.255.255.0
Router(config)#access-list 1 permit 192.168.0.0 0.0.0.255
Router(config)#ip nat inside source list 1 pool PUBLICO overload
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

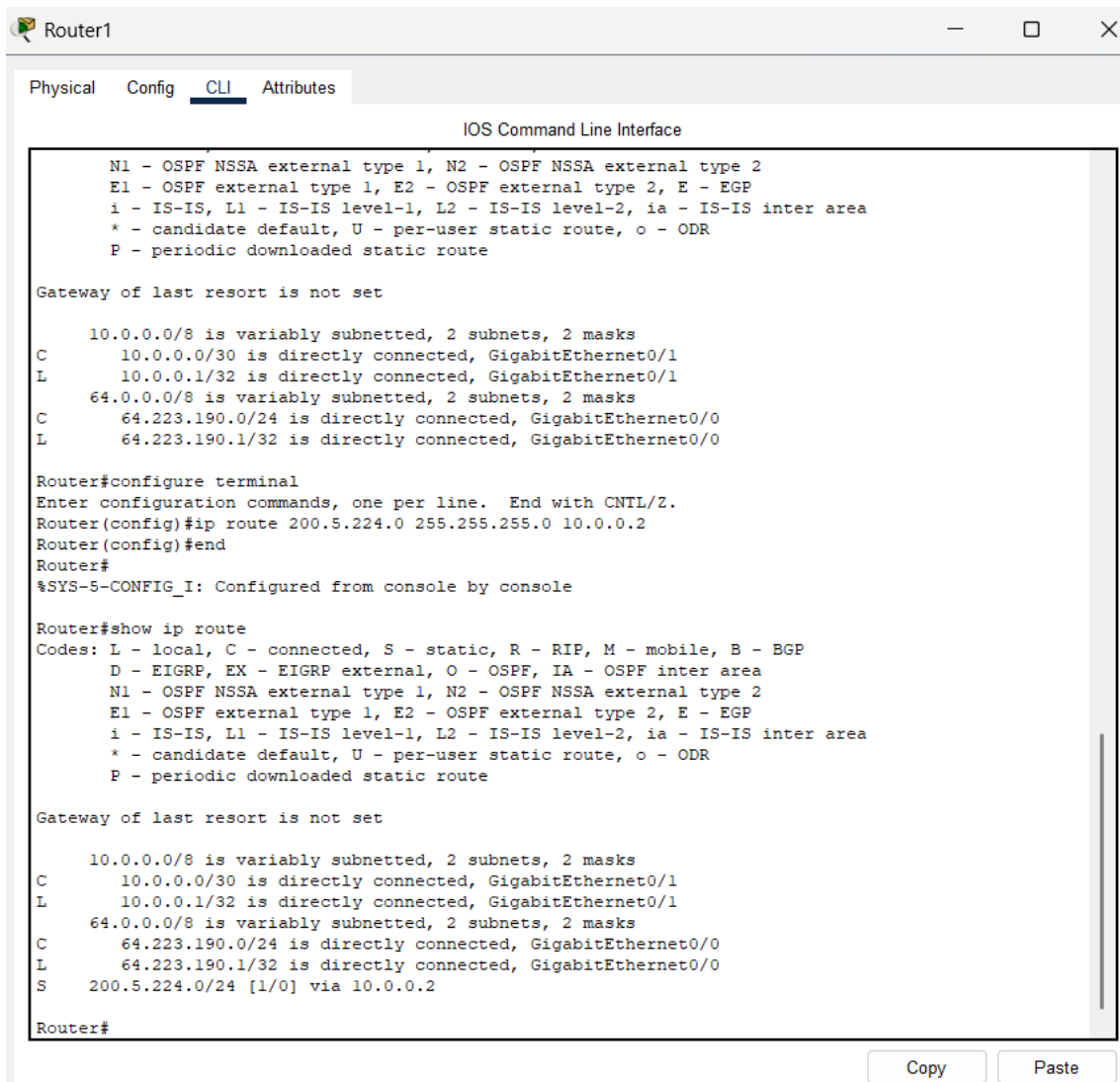
Copy Paste

Comandos de NAT cargados en Router0.

Ruta de retorno en Router1 (ISP)

Como el bloque 200.5.224.0/24 no está en ninguna interfaz, Router1 necesita una ruta estática para devolver el tráfico hacia Router0:

```
ip route 200.5.224.0 255.255.255.0 10.0.0.2
```



The screenshot shows a terminal window titled "Router1" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The output shows the configuration of static routes and the resulting routing table. The configuration includes setting up static routes for 200.5.224.0/24 via 10.0.0.2. The routing table shows the route as static (S) and connected (C) via 10.0.0.2.

```
Router1
Physical Config CLI Attributes

IOS Command Line Interface

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/1
L    10.0.0.1/32 is directly connected, GigabitEthernet0/1
64.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    64.223.190.0/24 is directly connected, GigabitEthernet0/0
L    64.223.190.1/32 is directly connected, GigabitEthernet0/0

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 200.5.224.0 255.255.255.0 10.0.0.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/1
L    10.0.0.1/32 is directly connected, GigabitEthernet0/1
64.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    64.223.190.0/24 is directly connected, GigabitEthernet0/0
L    64.223.190.1/32 is directly connected, GigabitEthernet0/0
S    200.5.224.0/24 [1/0] via 10.0.0.2

Router#
```

Router1: la ruta S 200.5.224.0/24 [1/0] via 10.0.0.2 aparece en la tabla.

NAT estático — MiWeb publicado en Internet

MiWeb debe ser accesible desde Internet por su IP pública 200.5.224.50.

Desde el servidor Google (fuera de la red) se navega a <http://200.5.224.50> y se accede a la página de MiWeb: el NAT estático traduce 200.5.224.50 → 192.168.0.15.

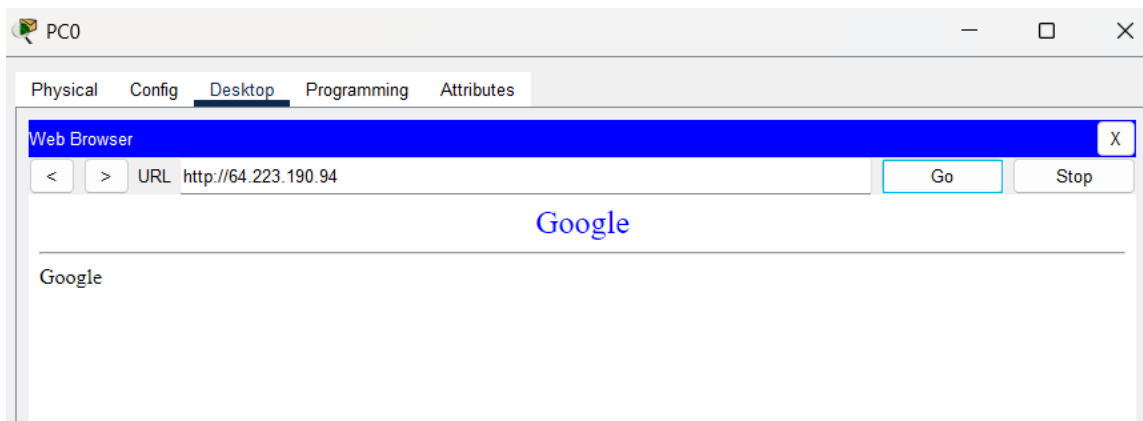


Google abre <http://200.5.224.50> → página de MiWeb ("Mi Pagina Web").

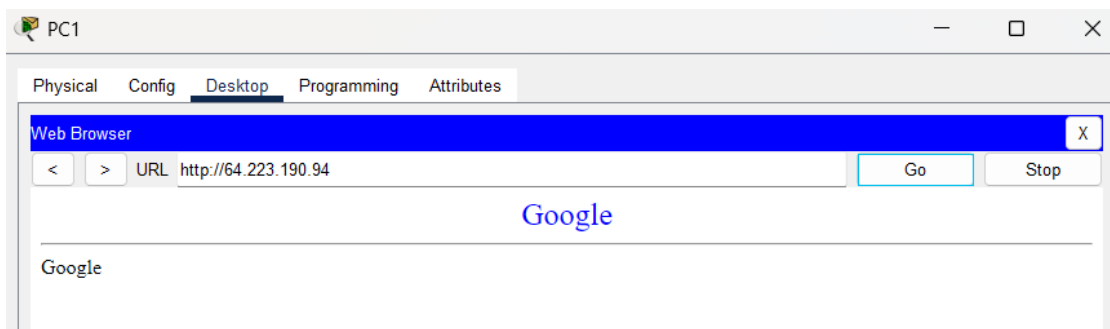
NAT dinámico con overload — navegación de las PCs

Las PCs internas navegan a Internet compartiendo el pool de IPs públicas.

PC0 y PC1 acceden a la página de Google (64.223.190.94). Ambas salen traducidas a la misma IP pública del pool (200.5.224.100), diferenciadas por el puerto de origen (PAT).



PC0 (192.168.0.10) navega a <http://64.223.190.94>.



PC1 (192.168.0.12) navega a <http://64.223.190.94>.

The screenshot shows a Cisco Router CLI window titled "Router0". The tabs at the top are "Physical", "Config", "CLI" (selected), and "Attributes". The main area displays the "IOS Command Line Interface".

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router>enable
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.0.1     YES manual up          up
GigabitEthernet0/1       10.0.0.2        YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.1
Router(config)#interface GigabitEthernet0/0
Router(config-if)# ip nat inside
Router(config-if)#interface GigabitEthernet0/1
Router(config-if)# ip nat outside
Router(config-if)#exit
Router(config)#
Router(config)#ip nat inside source static 192.168.0.15 200.5.224.50
Router(config)#ip nat pool PUBLICO 200.5.224.100 200.5.224.101 netmask 255.255.255.0
Router(config)#access-list 1 permit 192.168.0.0 0.0.0.255
Router(config)#ip nat inside source list 1 pool PUBLICO overload
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 200.5.224.100:1    192.168.0.10:1    64.223.190.94:1    64.223.190.94:1
icmp 200.5.224.100:2    192.168.0.10:2    64.223.190.94:2    64.223.190.94:2
icmp 200.5.224.100:3    192.168.0.10:3    64.223.190.94:3    64.223.190.94:3
icmp 200.5.224.100:4    192.168.0.10:4    64.223.190.94:4    64.223.190.94:4
icmp 200.5.224.100:5    192.168.0.10:5    64.223.190.94:5    64.223.190.94:5
icmp 200.5.224.100:6    192.168.0.10:6    64.223.190.94:6    64.223.190.94:6
icmp 200.5.224.100:7    192.168.0.10:7    64.223.190.94:7    64.223.190.94:7
icmp 200.5.224.100:8    192.168.0.10:8    64.223.190.94:8    64.223.190.94:8
--- 200.5.224.50        192.168.0.15      ---                ---
Router#
```

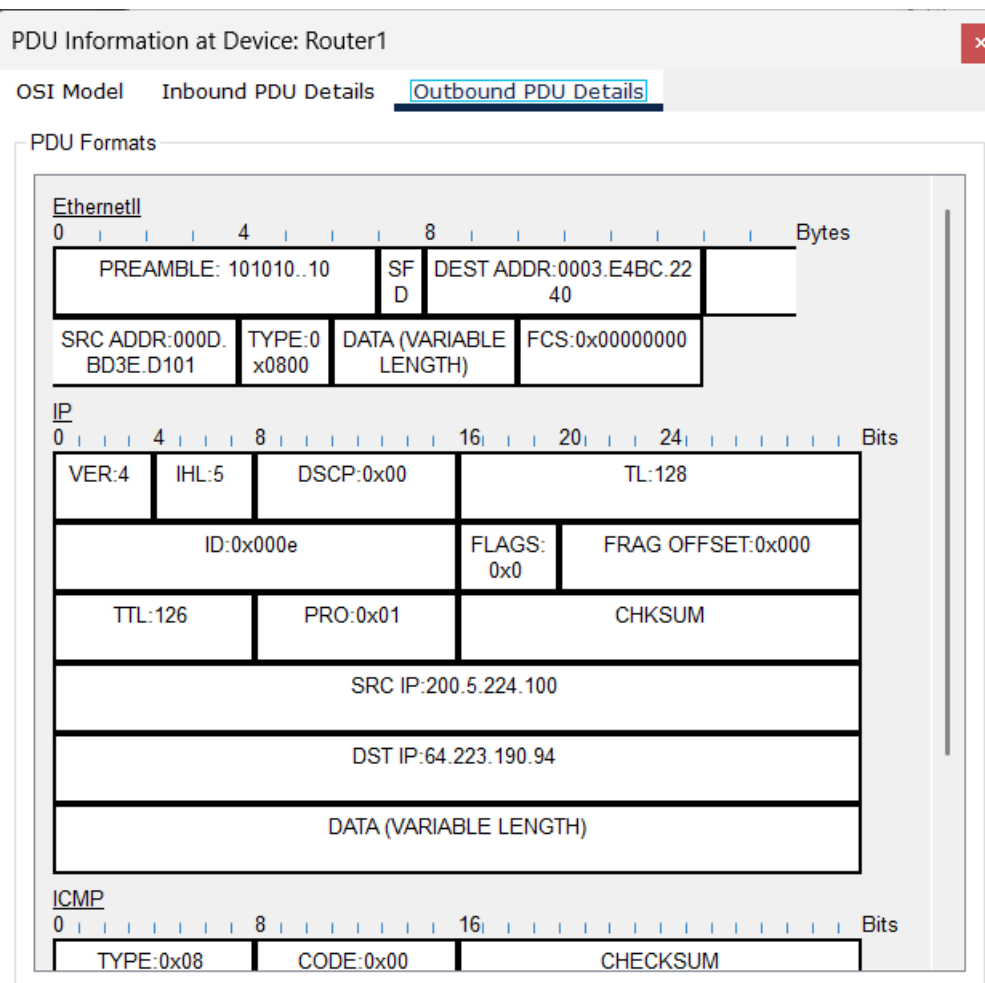
At the bottom right of the CLI window, there are "Copy" and "Paste" buttons.

show ip nat translations: el estático (--- 200.5.224.50 → 192.168.0.15) y el overload de la PC (200.5.224.100 con distintos puertos).

Transformación de las cabeceras (simulación)

Observe cómo se transforman las cabeceras de los paquetes.

En el paquete que sale hacia Google, la dirección IP de origen ya fue traducida: de la privada 192.168.0.10 a la pública 200.5.224.100. En overload también se modifica el puerto origen (TCP/UDP) para distinguir cada conexión.



PDU en Router1: SRC IP = 200.5.224.100 (traducida), DST IP = 64.223.190.94.

Conclusión

NAT permite que una red privada (RFC 1918) acceda a Internet usando direcciones públicas. El NAT estático mapea uno a uno una dirección interna a una pública (para publicar servidores, como MiWeb). El NAT dinámico con overload (PAT) permite que muchos hosts privados compartan pocas IPs públicas, diferenciando las conexiones por el puerto de origen. En ambos casos el router reescribe las cabeceras IP (y, en overload, los puertos TCP/UDP) al cruzar de la red interna a la externa.